

Clinical Risk Assessment Report: Patient 81

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Negative for Metastasis
- Prediction Confidence: Moderate Confidence
- Absolute Predicted Risk: 9.7% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 0.44x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as Low-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Small Zone Emphasis (PET) (GLZLM_SZE.PET), which reduced the patient's absolute risk by 1.4%. Biologically, this feature reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

2. Key Pathological & Imaging Drivers (Elevating Risk)

No major risk-elevating features observed in the top drivers.

3. Protective / Mitigating Factors (Reducing Risk)

Small Zone Emphasis (PET) (GLZLM_SZE.PET) **Value: -0.06 [Avg (46th %ile)] (-1.4%)**
Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

Large Zone Emphasis (CT) (GLZLM_LGZE.CT) **Value: -0.0 [Top 25%] (-1.0%)**
Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

Maximum Tumor Density (CONVENTIONAL_HUmax) **Value: -0.66 [Bottom 16%] (-1.0%)**
Reflects the most dense solid components or calcifications within the primary tumor lesion.

Large Zone Low Gray-Level Emphasis (CT) (GLZLM_LZLGE.CT) **Value: -0.43 [Bottom <1%] (-0.8%)**
Indicates the dominance of large regions with low attenuation or metabolism, pointing to extensive necrosis or ground-glass opacities.

Short Run Low Gray-Level Emphasis (CT) (GLRLM_SRLGE.CT) **Value: -0.02 [Top 22%] (-0.7%)**
Indicates fragmented areas of low density, often associated with diffuse micro-necrosis or loose tissue structure.

Tumor Sphericity (CT) (SHAPE_Sphericity(onlyFor3DROI)).CT) **Value: 1.29 [Top 10%] (-0.7%)**
Quantifies how closely the tumor resembles a perfect sphere; lower values indicate irregular, spiculated, or highly invasive margins.

Minimum Metabolic Activity (SUVmin) (CONVENTIONAL_SUVbwmin) **Value: -0.98 [Bottom 11%] (-0.7%)**
Indicates the areas of lowest metabolic activity within the tumor, which may represent necrotic or poorly perfused regions.

Long Run High Gray-Level Emphasis (PET) (GLRLM_LRHGE.PET) **Value: -0.79 [Bottom 3%] (-0.7%)**
Points to continuous, elongated bands of high-density or highly active tumor tissue.

Machine Learning Feature Attribution (SHAP Decision Path)

